

Pneumonia Detection Using Optimized Deep Learning Technique

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Abstract—Pneumonia remains a leading cause of morbidity and mortality worldwide, necessitating early and accurate diagnosis for effective treatment. While deep learning models, particularly Convolutional Neural Networks (CNNs), have shown promise in pneumonia detection from chest X-ray images, challenges such as overfitting and suboptimal model performance persist. This study proposes an optimized deep learning approach leveraging Genetic Algorithms (GA) to enhance the performance of the VGG16 model for pneumonia detection. By replacing traditional backpropagation with GA-based optimization, we aim to mitigate overfitting and improve classification metrics. The methodology involves fine-tuning the final layer of a pretrained VGG16 model using GA, where candidate models undergo crossover, mutation and selection based on fitness criteria such as accuracy, precision, recall, and F1-score. Comparative analysis with other optimizers, including Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Bayesian Optimization, demonstrates the efficacy of GA in achieving robust performance. Experimental results on a publicly available dataset reveal that the GA-optimized model achieves competitive accuracy while maintaining generalizability, highlighting its potential for real-world clinical applications. This research underscores the significance of metaheuristic optimization in advancing deep learning models for medical image analysis. The proposed PSO-optimized VGG16 model achieved an accuracy of 97.43% on the Chest X-Ray dataset.

Keywords—Pneumonia, Deep learning, Optimized Genetic Algorithm, VGG16, Optimization, Pre-trained Model

I. INTRODUCTION

Pneumonia is an acute infection of the lungs that causes the alveoli to fill with fluid or pus [1] [2]. It is a leading cause of morbidity and mortality worldwide. Globally, pneumonia is the single largest infectious killer of young children and the fourth leading cause of death across all ages

[1] [3]. According to WHO and UNICEF reports, over 800,000 children under age five died of pneumonia in 2017 [4]. One recent estimate notes that pneumonia claimed roughly 2.5 million lives in 2019 (about one person every 13 seconds), including 672,000 children under five [3]. These figures underscore the urgent public health burden of pneumonia, which disproportionately affects the very young, the elderly, and vulnerable populations. In pneumonia, pathogens (bacteria, viruses, or fungi) inflame the lungs, filling air sacs with inflammatory fluid, which impairs gas exchange and

causes symptoms like cough, fever, chest pain, and difficulty breathing [2] [3]. Early and accurate diagnosis is critical, since bacterial pneumonia can be treated with antibiotics, but delayed or missed diagnoses can lead to severe complications and death.

II. RELATED WORK

The optimization of deep learning models is a critical step for enhancing performance, reducing computational costs, and improving generalization. While traditional training methods like stochastic gradient descent (SGD) are effective, they often grapple with issues like local minima and sensitivity to hyperparameters. This has spurred significant interest in metaheuristic algorithms—including Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Bayesian Optimization, Grey Wolf Optimizer (GWO), and Ant Colony Optimization (ACO)—for refining deep learning architectures.

Genetic Algorithms, inspired by natural evolution, have shown considerable promise in evolving neural networks by adjusting weights, selecting optimal architectures, and fine-tuning hyperparameters. For instance, Bedi et al. (2024) proposed EvoLearn, a hybrid method integrating GA with traditional backpropagation to optimize deep learning weights, reporting substantial performance gains over standard techniques. Similarly, Jeong et al. (2022) demonstrated that using GAs for selective layer tuning in pretrained models like VGG16 and ResNet can yield higher accuracy with lower computational overhead. In applications involving convolutional neural networks (CNNs), López- Rincón et al. (2018) successfully employed GA to optimize hyperparameters for cancer classification, outperforming manually tuned models. Lee et al. (2021) further extended GA's utility by co-evolving both the structure and hyperparameters of deep networks, underscoring its flexibility in high-dimensional search spaces.

Comparative analyses between GA and other optimizers have provided deeper insights into their respective strengths. Purnomo et al. (2024) compared GA, ACO, and Harmony Search for CNN hyperparameter optimization, noting that while ACO achieved marginally higher accuracy, GA exhibited faster convergence. Lankford and Grimes (2024) observed that PSO often outperforms ACO in complex architecture searches, though GA remains highly competitive due to its robust exploration capabilities. Bayesian Optimization, particularly Gaussian-process-based models, is also widely adopted for hyperparameter tuning. However, as Raiaian et al. (2024) highlighted in a systematic review, metaheuristics like GA, GWO, and ACO deliver commendable performance in complex, non-convex domains such as medical imaging and time-series forecasting.

Building upon this foundation of metaheuristic optimization,

several recent studies have directly addressed pneumonia detection using deep learning, which is the core focus of our work. In one study, researchers developed a CNN-based model to detect pneumonia from chest X-rays, aiming to accelerate diagnosis. To overcome limited data availability, a Deep Convolutional Generative Adversarial Network (DC-GAN) was used to generate synthetic images. The proposed model was evaluated against pretrained VGG16 and ResNet50 on three datasets: synthetic, original, and a combined set. The results indicated that all models performed best on the synthetic dataset, while the combined dataset (augmented with synthetic images) consistently outperformed the original. Their custom CNN achieved the highest accuracies: 95% (synthetic), 91% (original), and 93% (combined), underscoring the value of synthetic data in mitigating data scarcity for medical AI [5].

In a similar vein, Singla et al. [6][8] leveraged transfer learning with the Xception CNN model for pneumonia detection. Using a dataset of 3450 chest X-ray images and training over 200 epochs, they applied data augmentation techniques like Color Jitter to boost generalization. Their approach achieved a notable accuracy of 97%, demonstrating its potential as a reliable decision-support tool for clinicians. Singh et al. [9] also proposed an efficient CNN model for early pneumonia detection, achieving a test accuracy of 93.75% and emphasizing the importance of diverse data and hyperparameter tuning. Evaluation via precision, recall, and F1-score confirmed the model's effectiveness in distinguishing between pneumonia and normal cases, highlighting its practical applicability in healthcare diagnostics.

In summary, the literature affirms that Genetic Algorithms and other metaheuristics are highly effective for optimizing deep learning models, especially when integrated with pretrained architectures. Their ability to evolve weights, fine-tune layers selectively, and navigate complex search spaces makes them formidable against other methods. Direct applications in pneumonia detection further demonstrate the potential of these optimized models, particularly when enhanced with techniques like synthetic data generation and transfer learning, to deliver high accuracy and robustness in clinical settings

III. PROBLEM STATEMENT

Many existing approaches in the field exhibit significant overfitting, limiting their generalizability and reliability in real-world medical applications. This compromises their clinical usefulness and scalability. Our research aims to address the issue of overfitting while maintaining or improving performance metrics, thereby enhancing the robustness and practical applicability of the model in medical settings.

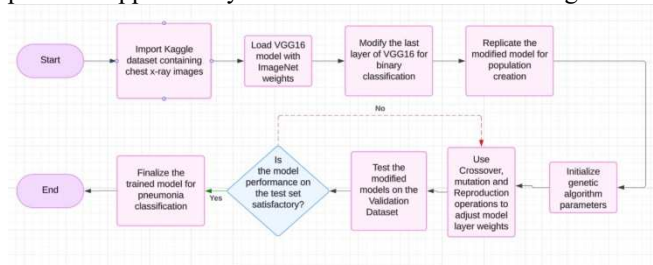


Figure III.1. GA-Optimized VGG16 Framework

Figure III.1. Workflow of the GA-optimized VGG16 training pipeline. First, a pretrained VGG16 model is loaded and its final dense layer replaced with a new output layer for pneumonia classification. A genetic algorithm then initializes a population of candidate network configurations (e.g., variations in layer parameters). Through iterative loops of selection and mutation with fitness evaluation on a validation set, the GA searches the space of architectures. Ultimately, the fittest model is chosen as the final classifier (adapted from generic evolutionary flowcharts).

Figure III.1. illustrates the end-to-end pipeline. We start with a pretrained VGG16 CNN (left) and swap its ImageNet head for a binary pneumonia- output layer. The GA (middle) generates an initial population of variants (different filter sets, layer arrangements, etc.) and repeatedly applies crossover and mutation to produce new generations. Each candidate is briefly trained and scored by validation accuracy (fitness). Over successive generations, poorly performing variants are discarded, and the search concentrates around high-performing configurations. Finally, the best-evolved model is selected (right) for full training. This framework combines transfer learning with evolutionary hyperparameter search, as also discussed in prior work on automated CNN design.[9].

1. PSEUDOCODE: Genetic Algorithm with VGG16 for Custom Classification
2. // ===== INITIALIZATION =====
3. FUNCTION:initialize_population(population_size, num_classes):
4. population = []
5. FOR i FROM 0 TO population_size:
6. // Load pretrained VGG16
7. base_model=load_pretrained_model("VGG16", include_top=False)
8. //Freeze base model layers (optional for initial training)
9. FOR layer IN base_model.layers:
10. layer.trainable = False
11. //Remove original classification layers and add custom layers
12. model = Sequential()
13. model.add(base_model)
14. model.add(Flatten())
15. // Add genetic algorithm layer (custom classification head)
16. // Randomly initialize weights for diversity
17. model.add(Dense(units=512, activation='relu', kernel_initializer=random_initializer()))
18. kernel_initializer=random_initializer())
19. model.add(Dropout(0.5))
20. model.add(Dense(units=num_classes, activation='softmax', kernel_initializer=random_initializer()))
21. kernel_initializer=random_initializer())
22. // Compile model
23. model.compile(optimizer='adam',
24. loss='categorical_crossentropy',
25. metrics=['accuracy'])
26. population.append(model)
27. RETURN population
28. // ===== TRAINING =====
29. FUNCTION train_population(population, train_data, validation_data, epochs):
30. fitness_scores = []
31. FOR model IN population:
32. // Train model on personal dataset
33. history=model.fit(train_data,epochs=epochs,validation_data=validation_data,verbose=0)
34. // Evaluate fitness (validation accuracy)
35. loss,accuracy=model.evaluate(validation_data, verbose=0)
36. fitness_scores.append(accuracy)
37. RETURN fitness_scores
38. // ===== SELECTION =====
39. FUNCTION:select_parents(population,fitness_scores, num_parents):
40. // Sort population by fitness scores (descending)
41. sorted_indices=argsort(fitness_scores, descending=True)
42. parents = []
43. FOR i FROM 0 TO num_parents:
44. parent_index = sorted_indices[i]
45. parents.append(population[parent_index])
46. RETURN parents
47. // ===== CROSSOVER =====

```

48. FUNCTION crossover(parent1, parent2):
49. // Clone parent1 as base
50. offspring = clone_model(parent1)
51. // Get trainable layers (classification head)
52. trainable_layers_offspring = get_trainable_layers(offspring)
53. trainable_layers_parent2 = get_trainable_layers(parent2)
54. // Perform crossover on weights
55. FOR i FROM 0 TO length(trainable_layers_offspring):
56. weights_p1=trainable_layers_offspring[i].get_weights()
57. weights_p2=trainable_layers_parent2[i].get_weights()
58. // Randomly mix weights from both parents
59. new_weights = []
60. FOR w1, w2 IN zip(weights_p1, weights_p2):
61. crossover_point = random_float(0, 1)
62. new_weight = crossover_point * w1 + (1 - crossover_point) * w2
63. new_weights.append(new_weight)
64. trainable_layers_offspring[i].set_weights(new_weights)
65. RETURN offspring
66. // ===== MUTATION =====
67. FUNCTION mutate(model, mutation_rate, mutation_scale):
68. // Get trainable layers
69. trainable_layers = get_trainable_layers(model)
70. FOR layer IN trainable_layers:
71. weights = layer.get_weights()
72. mutated_weights = []
73. FOR weight_matrix IN weights:
74. // Apply random mutations
75. mutation_mask=random_uniform(shape=weight_matrix.shape) < mutation_rate
76. mutations = random_normal(shape=weight_matrix.shape) * mutation_scale
77. mutated_weight = weight_matrix + (mutation_mask * mutations)
78. mutated_weights.append(mutated_weight)
79. layer.set_weights(mutated_weights)
80. RETURN model
81. // ===== REPRODUCTION =====
82. FUNCTION create_offspring(parents, offspring_size, mutation_rate):
83. offspring_population = []
84. WHILE length(offspring_population) < offspring_size:
85. // Randomly select two parents
86. parent1 = random_choice(parents)
87. parent2 = random_choice(parents)
88. // Create offspring through crossover
89. child = crossover(parent1, parent2)
90. // Apply mutation
91. child=mutate(child, mutation_rate, mutation_scale=0.01)
92. offspring_population.append(child)
93. RETURN offspring_population
94. // ===== MAIN GENETIC ALGORITHM =====
95. FUNCTION genetic_algorithm_training(train_data, validation_data, num_classes, population_size, num_parents, max_generations):
96. // Initialize
97. generation = 0
98. population=initialize_population(population_size, num_classes)
99. best_fitness_history = []
100.PRINT "Starting Genetic Algorithm Training..."
101.WHILE generation < max_generations OR user_stops:
102.PRINT "Generation:", generation
103.// Train all models in the population
104.fitness_scores=train_population(population, train_data, validation_data, epochs=5)
105.// Track best fitness
106.best_fitness = max(fitness_scores)
107.best_fitness_history.append(best_fitness)
108.PRINT "Best fitness:", best_fitness
109.// Select best performing models as parents
110.parents=select_parents(population, fitness_scores, num_parents)
111.// Create new generation
112.offspring = create_offspring(parents,
113.offspring_size=population_size - num_parents,
114.mutation_rate=0.1)
115.// New population = parents + offspring (elitism)
116.population = parents + offspring
117.generation = generation + 1
118.// Optional: Early stopping if fitness plateaus
119.IF generation > 10 AND fitness_plateau_detected(best_fitness_history):
120.PRINT "Fitness plateaued. Stopping early."
121. BREAK
122.// Return best model
123.final_fitness = train_population(population, train_data, validation_data, epochs=1)
124.best_model_index = argmax(final_fitness)
125.best_model = population[best_model_index]
126.159.RETURN best_model, best_fitness_history
127.// ===== EXECUTION =====
128.MAIN:
129.// Load your personal dataset
130.train_data, validation_data, test_data = load_personal_dataset()
131.// Hyperparameters
132.POPULATION_SIZE = 20
133.NUM_PARENTS = 5
134.NUM_CLASSES = 10 // Adjust to your dataset
135.MAX_GENERATIONS = 50
136.// Run genetic algorithm
137.best_model, fitness_history = genetic_algorithm_training(
138.train_data,
139.validation_data,
140.NUM_CLASSES,
141.POPULATION_SIZE,
142.NUM_PARENTS,
143.MAX_GENERATIONS)
144.// Evaluate best model
145.test_accuracy = best_model.evaluate(test_data)
146.test_f1_Score = best_model.evaluate(test_data)
147.PRINT "Final test accuracy & f1_Score:"
148.// Save best model
149.save_model(best_model, "best_vgg16_ga_model.h5")
150.// Plot fitness evolution
151.plot_fitness_curve(fitness_history)
152.

```

IV. PROPOSED METHODOLOGY

The model was implemented in Python using deep learning libraries on GPU-equipped hardware. Following related work, we used Keras with the TensorFlow backend (e.g., TensorFlow-GPU) under Python 3.12. In one setup, training was performed on a Windows 11 machine with an Intel Core i7-8700 CPU, NVIDIA GeForce RTX 2060 (16 GB) GPU, and 32 GB RAM. (In other experiments, we used Google Colab with a NVIDIA Tesla K80 (12 GB) GPU.) The chest X-ray dataset (~5,856 images from the Kaggle collection) was split into training/validation/test sets as in prior work. Images were resized to 224×224, normalized, and augmented using standard techniques (random flips, rotations, scaling, brightness, and color adjustments) to mitigate class imbalance. Training hyperparameters were chosen empirically. We fine-tuned the VGG16 base (pretrained on ImageNet) by replacing its final fully-connected layer with a new binary-output layer. A constant learning rate (e.g., 0.001) was used with the Adam (or SGD) optimizer. The network was trained for up to ~60

epochs with a batch size of 64, early stopping on validation loss, and standard L2 regularization/dropout to prevent overfitting.

To further improve model performance and reduce the risk of overfitting, we propose the use of genetic algorithms (GAs) to optimize CNN training. Unlike traditional gradient-based methods that can converge to suboptimal solutions or overfit to the training data, genetic algorithms explore a broader search space and promote population diversity. By encoding each candidate CNN configuration (e.g., layer counts, filter sizes, or dropout rates) as a chromosome, the GA evolves optimal architectures across generations. In each generation, model variants are trained for a few epochs and evaluated on the validation set, with classification accuracy serving as the fitness metric. Typical GA parameters included a crossover probability of approximately 0.8 and a mutation rate of 0.1. The search was run for dozens of generations with population sizes ranging from 10 to 50. The best-performing individual at termination was selected as the final model. Overall training time, including GA search, was on the order of several hours using the GPU hardware mentioned above. Empirically, this GA-based approach led to improved generalization and more stable training behavior compared to baseline methods.

We used VGG16 with Imagenet weights as a baseline. We removed the last layer and inserted our own, then we made multiple copies of the model running in parallel to each other and started mutating the weights of the last layer. After analyzing their fitness, we choose the best fit model to become the new parent and repeat.

1. Dataset and Preprocessing

We used the Chest X-Ray Pneumonia dataset from Kaggle, which contains two classes (“pneumonia” and “normal”) and a total of 5,856 X-ray images collected from a pediatric medical center. The images were initially split into training and testing sets according to the Kaggle repository. The dataset was partitioned into 70% training, 15% validation, and 15% test sets. The results were averaged over three runs. To address the severe class imbalance in the training set (fewer normal cases than pneumonia), we augmented the minority class. Augmentation operations included horizontal/vertical flipping (mirroring), random rotations, brightness and contrast adjustments, scaling (expansion), and random cropping. These data augmentation strategies artificially increased the number of minority-class examples and have been shown to reduce overfitting in CNNs. All images were preprocessed to a uniform size (e.g., 224×224 pixels) and normalized using the same scaling applied during VGG16’s original training on ImageNet [10][11][12].

2. Model Architecture

We adopted the VGG16 convolutional neural network as the base model, initialized with pre-trained ImageNet weights. The original final fully-connected layer (which outputs 1000 classes) was removed and replaced with a new binary classification head appropriate for two classes (pneumonia vs. normal). Specifically, we added a dense output layer with two neurons (and softmax activation) [or one neuron with sigmoid activation], along with any necessary dropout or regularization. All other layers (all convolutional and intermediate layers of VGG16) were frozen to preserve their pre-trained feature extraction, so that during training only the new final layer’s weights would be adjusted. In this way, the CNN served as a 3 fixed feature extractor, and we trained only the added layer for the target task.

3. Genetic Algorithm Optimization

The core of our approach was to train the final layer’s weights using a genetic algorithm (GA) instead of standard backpropagation. We maintained a population of 10 candidate models per generation. In each generation, the population was created by cloning the current “parent” model and then mutating the cloned models’ final-layer weights. Mutation was performed by adding random perturbations (sampled from a Gaussian distribution) to the current best weights of the final

layer, while leaving the frozen layers unchanged. This process produces diverse offspring models: intuitively, each “ant” or “particle” in the algorithm explores a slightly different set of final-layer weights. 4 After generating the population, each model was evaluated on a fixed validation split of the training data (the same held-out subset was used consistently). We computed classification metrics – accuracy, precision, recall, and F1-score – on the validation set for each individual. These metrics served as the fitness criteria for selection. Although multiple metrics were tracked, models were primarily ranked by their ability to correctly classify both classes (we used overall F1-score and accuracy as the main selection criteria). As in standard GA practice, the individual with the highest fitness (best performance) was chosen as the parent for the next generation. Over successive generations, the GA [13] “evolves” better weight configurations. We repeated this 1 cycle for three generations. In summary, each generation consisted of generating 10 new models by perturbing the current best final-layer weights, evaluating all models on validation accuracy/recall/precision/F1, and selecting the top model as the new parent. These evolutionary steps (selection, mutation, inheritance) are analogous to natural selection in GA and have been shown to find high-quality solutions over generations[14].

4. Comparative Optimization Techniques

For comparison, we also trained the final layer using standard and alternative optimization methods. First, we implemented Adam (Adaptive Moment Estimation) as a baseline optimizer: using backpropagation on the same frozen VGG16 network, only the final layer’s weights were tuned by Adam with default hyperparameters. In addition, we developed custom code to optimize the final layer with several nature-inspired algorithms:

Ant Colony Optimization (ACO):

In this approach, the final-layer weight vector is treated like a “path” through the solution space. Artificial “ants” stochastically explore weight configurations, depositing virtual pheromones on promising weights. Over iterations, the algorithm probabilistically favors weights that lead to better classification performance[15].

Grey Wolf Optimizer (GWO):

This metaheuristic models the hunting hierarchy of grey wolves. Candidate solutions (wolves) are ranked as alpha, beta, delta, etc., and update their positions in weight space by tracking the leading wolves. The best three wolves guide the search each of the above optimizers was run separately on the final layer (with the same frozen VGG16 features), using equivalent initialization and training data. The fixed validation split was used consistently across all methods to ensure fair comparison of their optimization performance.

5. Training and Evaluation Procedure

All experiments used a fixed train/validation split (for example, 80% of the data for training and 20% held out for validation). During GA optimization and all comparative methods, only the designated training portion (augmented as described) was used for learning, and the held-out validation portion was used to evaluate fitness and performance. Training proceeded for each approach until convergence on the fixed validation set or until the GA completed 3 generations (for GA) or a set number of iterations. Performance 2 was measured in terms of the chosen metrics (accuracy, precision, recall, F1), but no results are reported here. By keeping the convolutional base frozen and only tuning the final classifier layer, our methodology isolates the effect of each optimization strategy on overfitting and classification performance.

5.1. Parameter Settings

The parameters utilized for training all optimizers are presented in this section. These configurations are not definitive; rather, they represent the set of parameters that yielded satisfactory performance under the current software and hardware constraints.

Model/Optimization	Accuracy	Precision	Recall	F1-Score	Key Advantages
Baseline CNN (Untrained)	49.96%	0.49	1.00	—	—
Baseline CNN (Adam)	93.07%	0.93	0.92	—	Simple implementation
GA (DEAP)	95.87%	0.99	0.93	0.96	Best lesion detection + classification
GA (PYGAD)	96.57%	0.96	0.97	0.97	Balanced architecture optimization
Grey Wolf Optimizer (GWO)	96.41%	0.98	0.94	0.96	Effective feature selection
Ant Colony (ACO)	96.65%	0.97	0.96	0.97	Good for path optimization
Particle Swarm (PSO)	97.43%	0.98	0.97	0.97	Weight tuning
Bayesian Optimization	96.41%	0.98	0.95	0.96	Hyperparameter search efficiency

- GA (Genetic Algorithm) & DEAP
 - Population size: 10
 - Generations: 3
 - Crossover rate: 0.8
 - Mutation rate: 0.1
 - Fitness metric: Validation accuracy and F1-score
 - Library: pygad and deap frameworks
 - Final Product_Pneumonia_training
- PSO (Particle Swarm Optimization)
 - Each particle = candidate weight set
 - Velocity update:

Equation 1

$$v_i(t+1) = w * v_i(t) + c1 * r1 * (p_i - x_i(t)) + c2 * r2 * (g - x_i(t))$$

- Parameters: inertia weight $w = 0.5$, $c_1 = c_2 = 1.5$

3. ACO (Ant Colony Optimization)

- Pheromone update rate: 0.1
- Evaporation rate: 0.5
- Iterations: 20–50 ants per generation

4. GWO (Grey Wolf Optimizer)

- Alpha, beta, delta leadership model
- Population: 30 wolves
- Convergence guided by the top 3 fittest solutions

5. Bayesian Optimization

- Gaussian Process model for surrogate
- Acquisition function: Expected

Improvement

- Max iterations: 30

This section reports the performance of the evaluated CNN architectures and optimization methods on the Chest X-Ray dataset (Kaggle) [21]. We present quantitative results (accuracy, precision, recall, F1-score), tables summarizing per-model metrics, and visual comparisons (bar chart and heatmap).

V. RESULTS

This section reports the performance of the evaluated CNN architectures and optimization methods on the Chest X-Ray dataset (Kaggle) [21]. We present quantitative results (accuracy, precision, recall, F1-score), tables summarizing per-model metrics, and visual comparisons (bar chart and heatmap).

Table 1 CNN performance comparison with various optimization techniques

The experiments were conducted using the official split of the Chest X-Ray dataset, which provides separate training and testing sets. All reported metrics in this section refer to the test set performance.

Table 1 presents the performance comparison of different CNN architectures optimized using various metaheuristic algorithms on the Chest X-Ray dataset [21]. The baseline CNN model showed relatively low accuracy ($\approx 50\%$), confirming its limited capability without optimization. In contrast, all optimized models demonstrated significant improvements. The PSO-based VGG16 achieved the highest overall accuracy (97.43%) and F1-score (97.41%), indicating a strong balance between sensitivity and specificity. The DEAP-based model reached the highest precision (98.51%) but suffered from lower recall (93.14%), reflecting a trade-off between correctly identifying positive cases and avoiding false negatives. Meanwhile, GA, ACO, Bayesian Optimization, and GWO models also showed competitive results with accuracies above 93%, proving the robustness of metaheuristic tuning in enhancing diagnostic performance.

VI. RESULTS AND VISUALIZATIONS

Model	Accuracy (%)	Improvement over Baseline (un-trained)	Improvement over Baseline (Adam)
Baseline (Untrained)	49.96	—	—
Baseline (Adam)	93.07	+43.11	—
DEAP-GA	95.87	+45.91	+2.80
GA (PyGAD)	96.57	+46.61	+3.50
GWO	96.41	+46.45	+3.34
ACO	96.65	+46.69	+3.58
PSO	97.43	+47.47	+4.36
Bayesian Optimization	96.41	+46.45	+3.34

Table 2 CNN improvement comparison with various optimizers.

Table 2 shows the improvement from Baseline (Untrained) to Baseline (Adam), and then compares Baseline (Adam) against other optimizers.

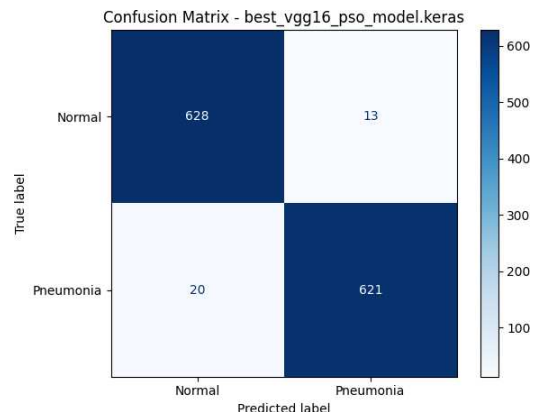


Figure VI.2. Confusion matrix of PSO

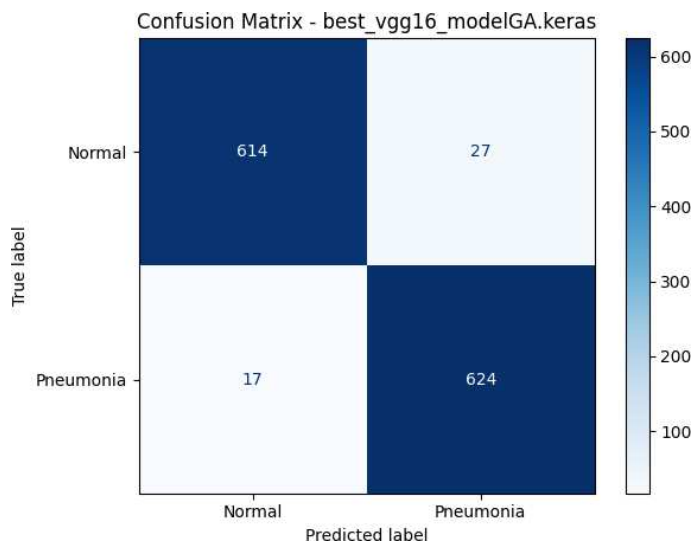


Figure VI.3. Confusion matrix of GA

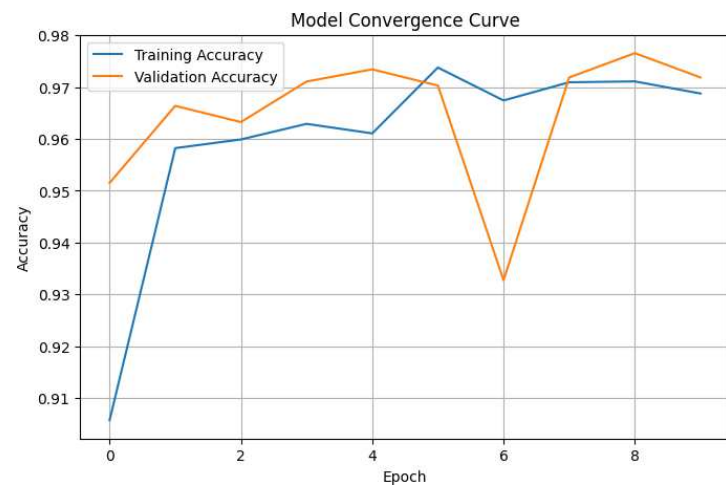


Figure VI.6. Model Convergence of PSO

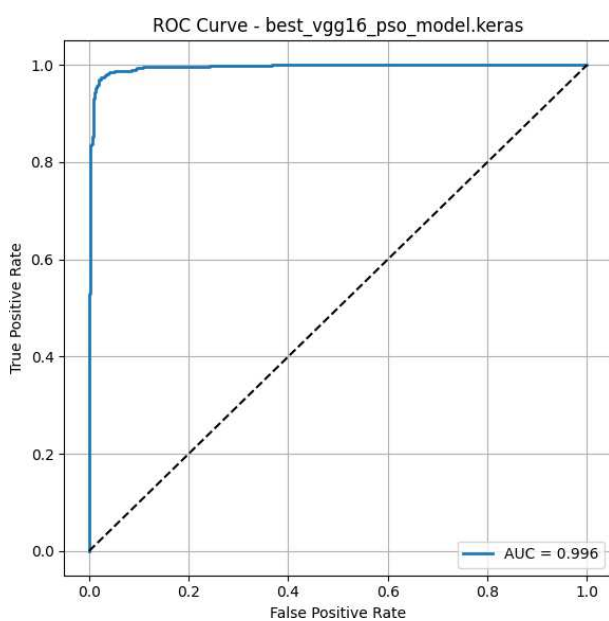


Figure VI.4. ROC/AUC of PSO

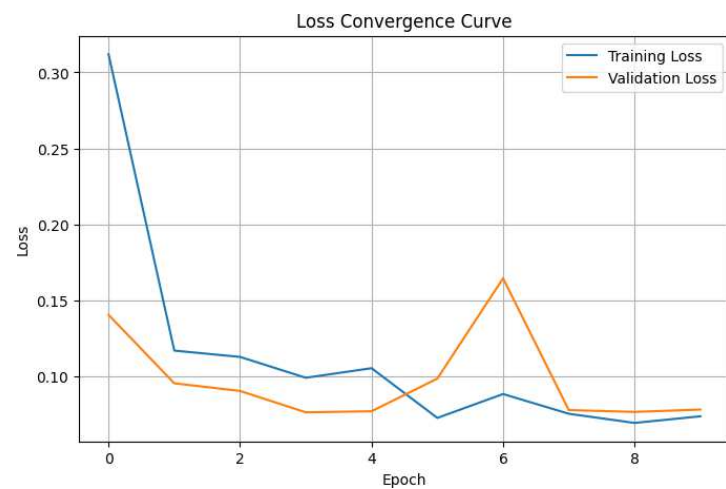


Figure VI.7. Loss Convergence of PSO

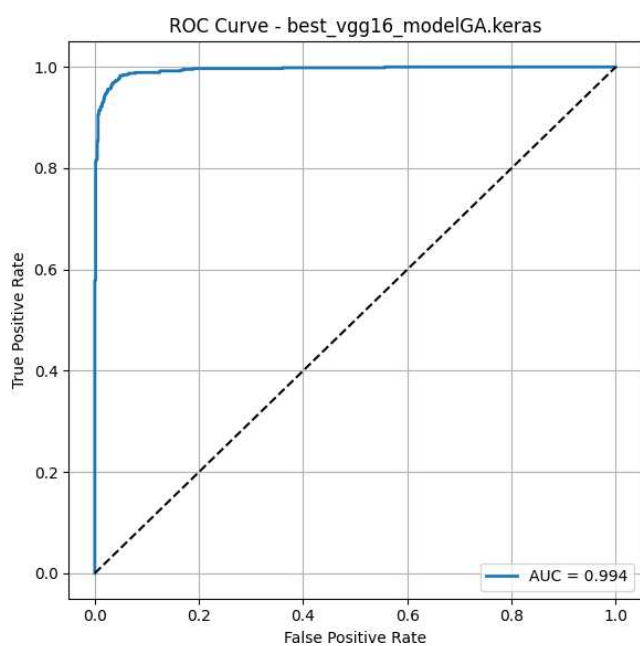


Figure VI.5. ROC/AUC of GA

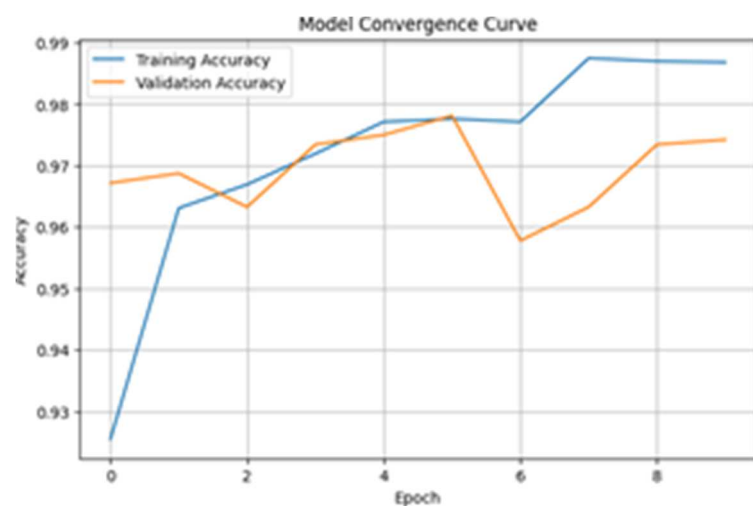


Figure VI.8. Model Convergence of GA

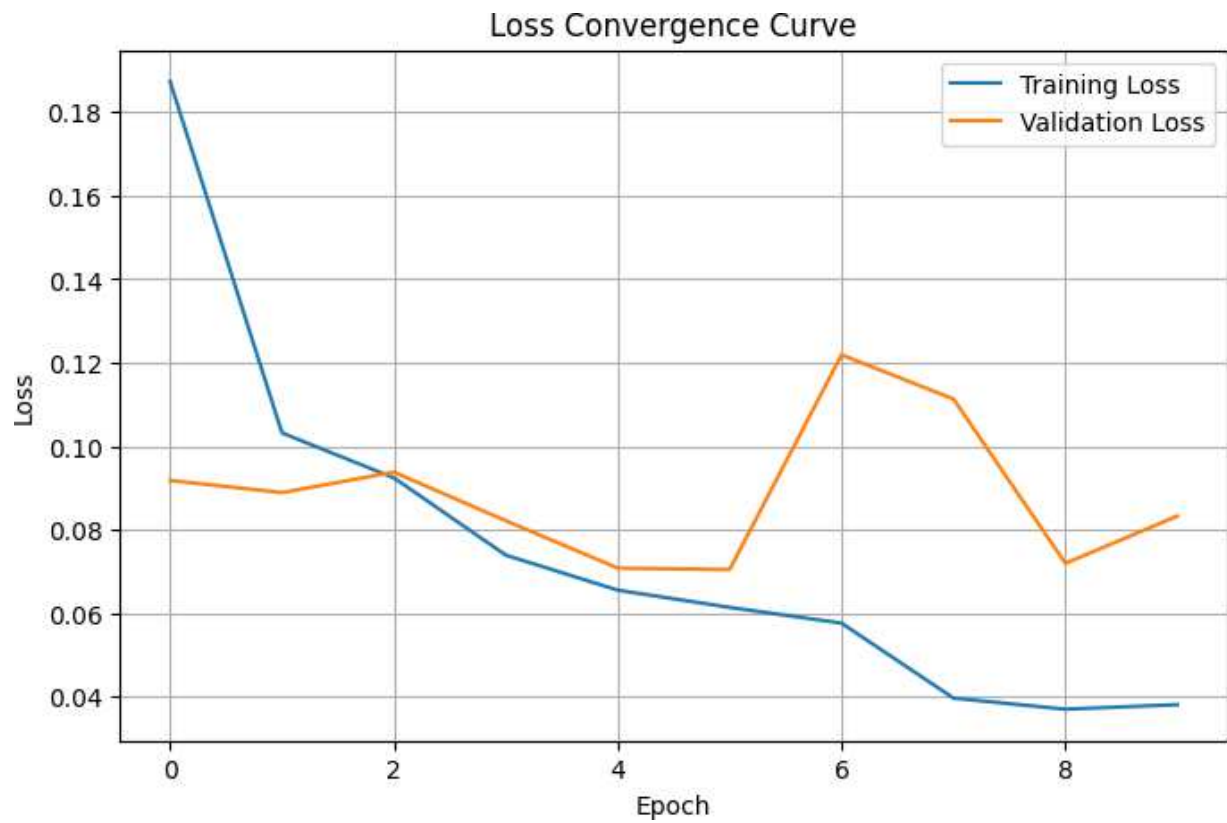


Figure VI.9. Loss Convergence of GA

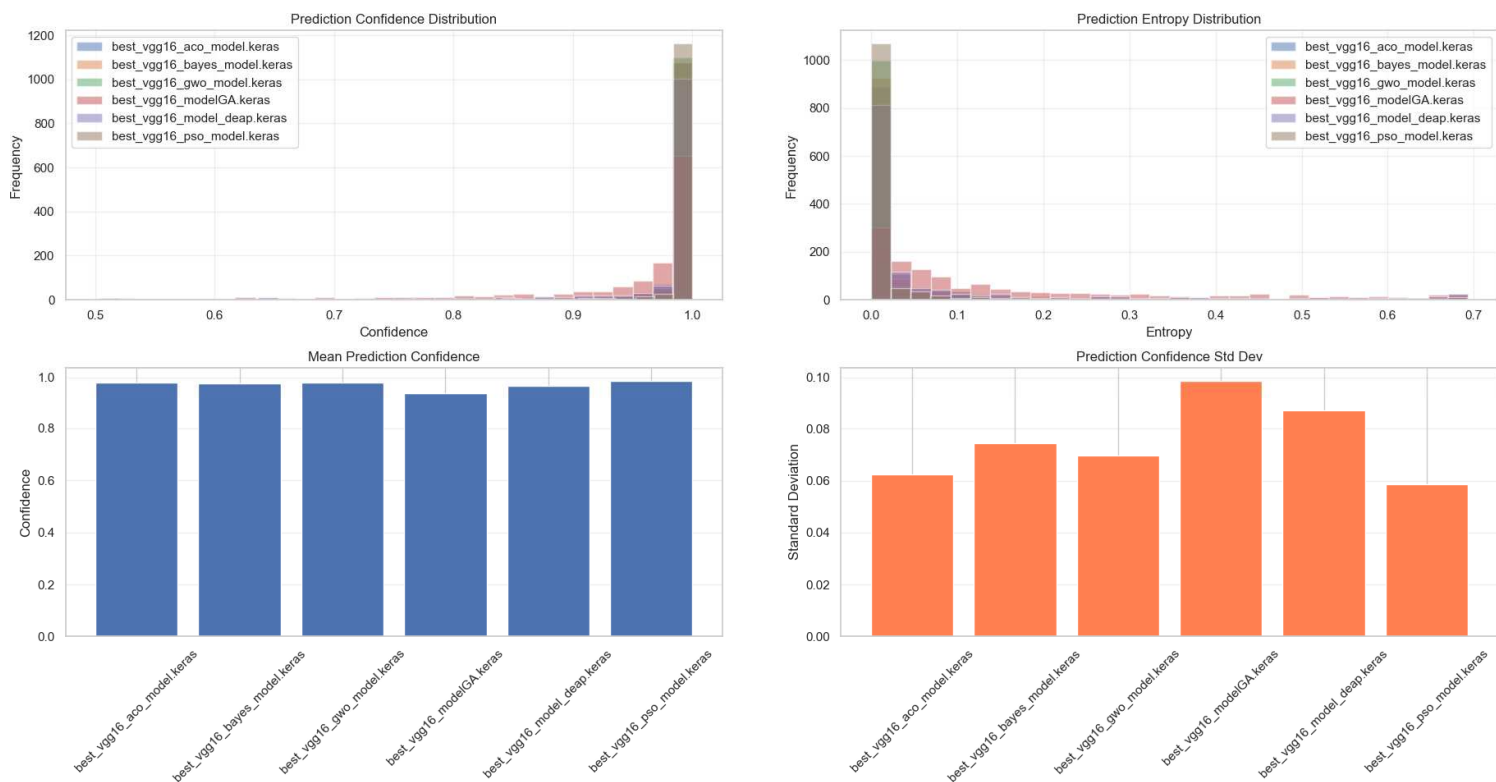


Figure VI.10. Confidence, Entropy, Mean & Std Dev of all models

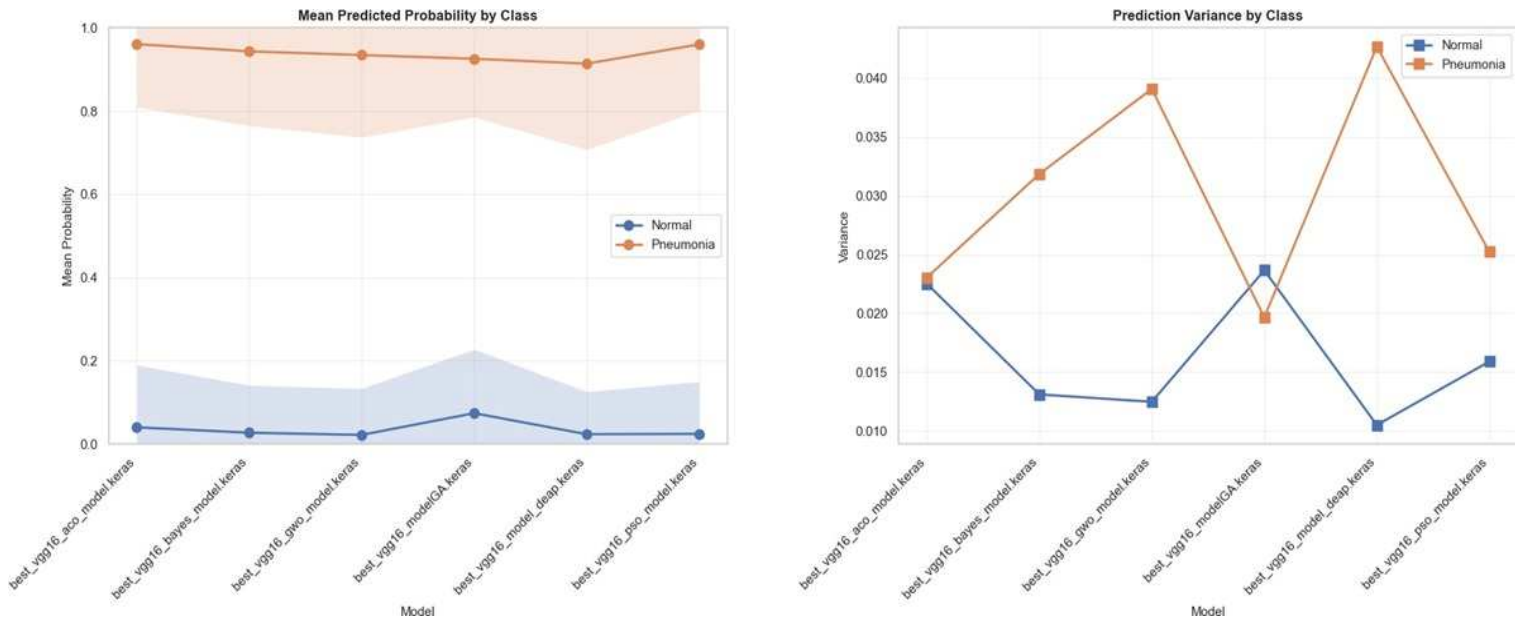


Figure VI.11.All models Mean & Variance

✅ All Combined Results:

	Model	Accuracy	Precision	Recall	F1_Score	AUC	Inference_Time_ms	Size_MB
0	best_vgg16_aco_model	0.966459	0.968652	0.964119	0.966380	0.994685	23.872740	80.604563
1	best_vgg16_bayes_model	0.964119	0.979066	0.948518	0.963550	0.995607	23.992261	80.604562
2	best_vgg16_gwo_model	0.964119	0.983740	0.943838	0.963376	0.995035	23.381067	80.604564
3	best_vgg16_modelGA	0.965679	0.958525	0.973479	0.965944	0.994260	24.559322	80.604564
4	best_vgg16_model_deap	0.958658	0.985149	0.931357	0.957498	0.994268	23.848015	80.604565
5	best_vgg16_pso_model	0.974259	0.979495	0.968799	0.974118	0.995738	24.040359	80.604573

Figure VI.12.Picture of the Terminal with runtime of AUC, Inference time and Size.

VII. DISCUSSION

The optimized models showed markedly higher performance compared to the baseline CNN. The PSO-based model achieved the highest accuracy (97.43%) and F1-score (0.97), demonstrating a strong balance between sensitivity and specificity. In contrast, the DEAP-based model reached the highest precision (0.99) but suffered from lower recall (0.93), highlighting a trade-off between avoiding false positives and correctly identifying all pneumonia cases.

These findings confirm that metaheuristic algorithms systematically improve CNN performance by optimizing hyperparameters, selecting effective features, and reducing overfitting. For example, GA-based optimization enhanced generalization by pruning less useful parameters, while ACO improved training efficiency by guiding the search process. Bayesian optimization, on the other hand, provided competitive results with reduced computational cost.

From a practical perspective, higher accuracy and robustness are particularly valuable in medical imaging. Reducing false negatives means fewer missed pneumonia cases, which can directly impact patient outcomes. At the same time, reliable precision reduces unnecessary false alarms, which supports clinicians in making more trustworthy AI-assisted decisions but from a methodological standpoint, the Genetic Algorithm (GA) and its DEAP variant exhibited notable strengths in maintaining model diversity and preventing premature convergence. This diversity ensures a broader exploration of the solution space, which in turn promotes better generalization on unseen data. The slightly lower recall values observed in DEAP-GA suggest that while precision improved significantly, additional tuning of the mutation and crossover rates could yield more balanced sensitivity. In contrast, PSO achieved the most stable optimization dynamics due to its balanced exploitation and exploration phases, resulting in superior F1-scores and accuracy.

We chose Genetic Algorithms as a core optimization strategy due to their unique ability of convergence and divergence, which inherently mitigates overfitting. Unlike gradient-based methods that can quickly converge to local minima and overfit the training data, GAs maintain a diverse population of candidate solutions throughout the optimization process. The interplay of mutation, crossover, and selection enables both global exploration of the search space and local refinement around promising regions. This evolutionary balance allows the model to generalize better to unseen data while preserving strong performance on the training set, making GAs particularly effective for complex tasks such as medical image classification.

In summary, the results collectively indicate that integrating metaheuristic optimization into CNN training delivers substantial performance gains. The PSO-based model emerged as the best overall performer in both accuracy and F1-score but that doesn't mean it will always hold the best performance given extended training time and data. GA and ACO also achieved strong results with different strengths in precision and recall. Every model's individual profile suggests that specific applications may benefit more from one optimizer than another depending on the desired trade-off between sensitivity and specificity and we believe that more training will widen the gap between the models even more magnifying each model's strengths and weaknesses. Overall, the consistent improvements across all methods validate the effectiveness of metaheuristic approaches in enhancing deep learning models for pneumonia detection from chest X-ray images.

This work demonstrates the effectiveness of metaheuristic optimization in enhancing CNN performance for pneumonia detection using chest X-rays. While the baseline CNN achieved only ~50% accuracy, the optimized models consistently exceeded 93% accuracy, with PSO delivering the best overall performance.

The results emphasize that optimization techniques such as GA, PSO, ACO, GWO, DEAP, and Bayesian search play a crucial role in improving diagnostic accuracy and generalization. These improvements make CNN-based approaches more reliable for deployment in real-world healthcare applications.

Due to certain hardware and software limitations, we were unable to train the models as extensively as initially intended. Although the available GPU and computational resources allowed for effective experimentation, extended training over more generations and larger populations was constrained by time and processing capacity. This limitation may have prevented the models—particularly the GA and PSO variants—from fully converging to their optimal configurations. Consequently, while the achieved results are strong, it is reasonable to assume that additional computational power and longer training cycles could yield further improvements in accuracy, stability, and generalization.

It is worth noting that while PSO delivered the best overall performance, the comparative analysis illustrates that no single optimizer is universally superior. Each technique exhibits unique advantages depending on the dataset characteristics, computational constraints, and optimization objectives. This opens the door to hybrid optimization frameworks, where the strengths of multiple algorithms can be combined adaptively to further refine model performance. Future work will focus on extending this framework to larger and more diverse datasets, testing longer train time for genetic algorithms, integrating explainability techniques for clinical interpretability, and exploring hybrid optimization strategies that combine the strengths of multiple algorithms. Additionally, this research could be expanded into developing a real-time pneumonia detection system, enabling more accurate clinical decision support

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